

# Exercise 2A Step-by-Step Solutions

## Comprehensive Guide to Fractional Arithmetic and Operations

Based on the Chapter 2: Fractions Source Materials

### Mathematical Foundations & Rules

Before solving the exercises, we summarize the core mathematical rules and principles of fractional operations established in the source materials:

- 1. Product of Fractions:** The product of two or more fractions is defined as:

$$\text{Product of Fractions} = \frac{\text{Product of their Numerators}}{\text{Product of their Denominators}} \implies \frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

Where  $a, b, c, d$  are natural numbers.

- 2. Multiplicative Inverse (Reciprocal):** If two fractions are such that their product is 1, then each is called the reciprocal of the other. The reciprocal of a non-zero fraction  $\frac{a}{b}$  is  $\frac{b}{a}$ , and vice-versa.

- 3. Fraction as an Operator 'of':** The word 'of' works exactly like multiplication. Thus, we define:

$$\frac{a}{b} \text{ of } \frac{c}{d} = \frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

- 4. Mixed Fractions:** Before performing multiplication or division, mixed numbers must always be converted into improper fractions:

$$w\frac{a}{b} = \frac{(w \times b) + a}{b}$$

## Section 1: Find the Product

Evaluate and simplify each of the following products.

### Question 1

Find the product:  $\frac{5}{6} \times \frac{3}{7}$

$$\text{Product} = \frac{5 \times 3}{6 \times 7}$$

Since 3 in the numerator and 6 in the denominator share a common factor of 3, we simplify:

$$\begin{aligned} \text{Product} &= \frac{5 \times 1}{2 \times 7} \\ &= \frac{5}{14} \end{aligned}$$

### Question 2

Find the product:  $\frac{7}{18} \times \frac{9}{14}$

$$\text{Product} = \frac{7 \times 9}{18 \times 14}$$

We can cross-cancel the common factors:

$$\begin{aligned} \text{Numerator 7 and Denominator 14} &\implies \frac{7}{14} = \frac{1}{2} \\ \text{Numerator 9 and Denominator 18} &\implies \frac{9}{18} = \frac{1}{2} \\ \text{Product} &= \frac{1 \times 1}{2 \times 2} \\ &= \frac{1}{4} \end{aligned}$$

### Question 3

Find the product:  $28 \times \frac{7}{8}$

$$\text{Product} = \frac{28}{1} \times \frac{7}{8} = \frac{28 \times 7}{8}$$

Since 28 and 8 share a common factor of 4 ( $28 \div 4 = 7$  and  $8 \div 4 = 2$ ):

$$\text{Product} = \frac{7 \times 7}{2} = \frac{49}{2}$$

Converting the improper fraction back to a mixed number:

$$= 24\frac{1}{2}$$

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**Question 4**

Find the product:  $7 \times \frac{1}{7}$

$$\text{Product} = \frac{7}{1} \times \frac{1}{7} = \frac{7 \times 1}{1 \times 7}$$

The factors of 7 cancel out completely:

$$\begin{aligned} &= \frac{1}{1} \\ &= \mathbf{1} \end{aligned}$$

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**Question 5**

Find the product:  $2\frac{1}{25} \times \frac{5}{17}$

First, convert the mixed fraction  $2\frac{1}{25}$  to an improper fraction:

$$\begin{aligned} 2\frac{1}{25} &= \frac{2 \times 25 + 1}{25} = \frac{51}{25} \\ \text{Product} &= \frac{51}{25} \times \frac{5}{17} \end{aligned}$$

Simplify by cross-cancelling common factors ( $51 = 17 \times 3$  and  $25 = 5 \times 5$ ):

$$\begin{aligned} \text{Product} &= \frac{3 \times 1}{5 \times 1} \\ &= \frac{\mathbf{3}}{\mathbf{5}} \end{aligned}$$

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**Question 6**

Find the product:  $1\frac{1}{13} \times 7\frac{3}{7}$

First, convert the mixed fractions to improper fractions:

$$\begin{aligned} 1\frac{1}{13} &= \frac{1 \times 13 + 1}{13} = \frac{14}{13} \\ 7\frac{3}{7} &= \frac{7 \times 7 + 3}{7} = \frac{52}{7} \\ \text{Product} &= \frac{14}{13} \times \frac{52}{7} \end{aligned}$$

Simplify by cross-cancelling common factors ( $14 = 7 \times 2$  and  $52 = 13 \times 4$ ):

$$\begin{aligned} \text{Product} &= \frac{2 \times 4}{1 \times 1} \\ &= \mathbf{8} \end{aligned}$$

**Question 7**

Find the product:  $\frac{4}{17} \times 7\frac{1}{12}$

First, convert the mixed fraction  $7\frac{1}{12}$  to an improper fraction:

$$7\frac{1}{12} = \frac{7 \times 12 + 1}{12} = \frac{85}{12}$$

$$\text{Product} = \frac{4}{17} \times \frac{85}{12}$$

Simplify by cross-cancelling (4 and 12 share factor of 4;  $85 = 17 \times 5$ ):

$$\text{Product} = \frac{1 \times 5}{1 \times 3}$$

$$= \frac{5}{3}$$

Converting back to a mixed number:

$$= 1\frac{2}{3}$$

**Question 8**

Find the product:  $7\frac{1}{4} \times \frac{7}{58} \times 1\frac{11}{21}$

First, convert the mixed fractions to improper fractions:

$$7\frac{1}{4} = \frac{7 \times 4 + 1}{4} = \frac{29}{4}$$

$$1\frac{11}{21} = \frac{1 \times 21 + 11}{21} = \frac{32}{21}$$

$$\text{Product} = \frac{29}{4} \times \frac{7}{58} \times \frac{32}{21}$$

Grouping and cross-cancelling the terms:

$$\frac{29}{58} = \frac{1}{2} \quad (\text{since } 58 = 29 \times 2)$$

$$\frac{7}{21} = \frac{1}{3} \quad (\text{since } 21 = 7 \times 3)$$

$$\frac{32}{4} = 8$$

$$\text{Product} = 8 \times \frac{1}{2} \times \frac{1}{3}$$

$$= \frac{8}{6} = \frac{4}{3}$$

Converting to a mixed number:

$$= 1\frac{1}{3}$$

**Question 9**

Find the product:  $\frac{1}{19} \times 91 \times 5\frac{11}{13}$

First, convert the mixed fraction  $5\frac{11}{13}$  to an improper fraction:

$$5\frac{11}{13} = \frac{5 \times 13 + 11}{13} = \frac{76}{13}$$

$$\text{Product} = \frac{1}{19} \times \frac{91}{1} \times \frac{76}{13}$$

Simplify using common factors ( $91 \div 13 = 7$  and  $76 \div 19 = 4$ ):

$$\begin{aligned} \text{Product} &= 1 \times \frac{91}{13} \times \frac{76}{19} \\ &= 1 \times 7 \times 4 \\ &= \mathbf{28} \end{aligned}$$

**Question 10**

Find the product:  $7\frac{1}{2} \times 2\frac{3}{16} \times 2\frac{2}{7}$

First, convert all mixed fractions to improper fractions:

$$7\frac{1}{2} = \frac{7 \times 2 + 1}{2} = \frac{15}{2}$$

$$2\frac{3}{16} = \frac{2 \times 16 + 3}{16} = \frac{35}{16}$$

$$2\frac{2}{7} = \frac{2 \times 7 + 2}{7} = \frac{16}{7}$$

$$\text{Product} = \frac{15}{2} \times \frac{35}{16} \times \frac{16}{7}$$

Simplify by cancelling the 16 in numerator and denominator, and dividing 35 by 7:

$$\begin{aligned} \text{Product} &= \frac{15}{2} \times 5 \times 1 \\ &= \frac{75}{2} \end{aligned}$$

Converting back to a mixed number:

$$= \mathbf{37\frac{1}{2}}$$

## Section 2: Find the Value Of

Evaluate each expression by translating the operator ‘of’ as multiplication.

### Question 11 (i)

Find the value of:  $\frac{3}{4}$  of  $\frac{8}{9}$

$$\begin{aligned}\text{Value} &= \frac{3}{4} \times \frac{8}{9} \\ &= \frac{3}{9} \times \frac{8}{4} \\ &= \frac{1}{3} \times 2 \\ &= \frac{2}{3}\end{aligned}$$

### Question 11 (ii)

Find the value of:  $\frac{1}{2}$  of  $2\frac{2}{3}$

Convert the mixed fraction  $2\frac{2}{3}$ :

$$\begin{aligned}2\frac{2}{3} &= \frac{2 \times 3 + 2}{3} = \frac{8}{3} \\ \text{Value} &= \frac{1}{2} \times \frac{8}{3} \\ &= \frac{8}{6} = \frac{4}{3} \\ &= 1\frac{1}{3}\end{aligned}$$

### Question 11 (iii)

Find the value of:  $\frac{4}{5}$  of 1 hour

Since 1 hour = 60 minutes:

$$\begin{aligned}\text{Value} &= \frac{4}{5} \times 60 \text{ minutes} \\ &= 4 \times 12 \text{ minutes} \\ &= \mathbf{48} \text{ minutes}\end{aligned}$$

**Question 11 (iv)**

Find the value of:  $\frac{3}{5}$  of Rs. 1

Since Rs. 1 = 100 paise:

$$\begin{aligned}\text{Value} &= \frac{3}{5} \times 100 \text{ paise} \\ &= 3 \times 20 \text{ paise} \\ &= \mathbf{60} \text{ paise} \quad (\text{or Rs. } 0.60)\end{aligned}$$

**Question 11 (v)**

Find the value of:  $\frac{8}{15}$  of  $1\frac{2}{3}$  metres

Convert the mixed fraction  $1\frac{2}{3}$ :

$$\begin{aligned}1\frac{2}{3} &= \frac{1 \times 3 + 2}{3} = \frac{5}{3} \text{ metres} \\ \text{Value} &= \frac{8}{15} \times \frac{5}{3} \text{ metres}\end{aligned}$$

Simplify by cross-cancelling ( $5 \div 15 = 1 \div 3$ ):

$$\begin{aligned}&= \frac{8 \times 1}{3 \times 3} \\ &= \frac{\mathbf{8}}{\mathbf{9}} \text{ metres}\end{aligned}$$

**Question 11 (vi)**

Find the value of:  $\frac{5}{7}$  of  $2\frac{1}{3}$  kg

Convert the mixed fraction  $2\frac{1}{3}$ :

$$\begin{aligned}2\frac{1}{3} &= \frac{2 \times 3 + 1}{3} = \frac{7}{3} \text{ kg} \\ \text{Value} &= \frac{5}{7} \times \frac{7}{3} \text{ kg}\end{aligned}$$

The factors of 7 cancel out completely:

$$= \frac{\mathbf{5}}{\mathbf{3}} \text{ kg} = \mathbf{1}\frac{\mathbf{2}}{\mathbf{3}} \text{ kg}$$

## Section 3: Word Problems

Detailed solutions to the mathematical application problems in Exercise 2A.

### Question 12

A car can travel  $12\frac{1}{2}$  km in 1 litre of petrol. How much distance can it travel in  $42\frac{3}{5}$  litres of petrol?

**Solution:**

- Distance travelled with 1 litre of petrol =  $12\frac{1}{2}$  km =  $\frac{25}{2}$  km.
- Quantity of petrol available =  $42\frac{3}{5}$  litres =  $\frac{213}{5}$  litres.

The total distance the car can travel is the product of these two values:

$$\begin{aligned}\text{Distance} &= \text{Distance per litre} \times \text{Total petrol} \\ &= \frac{25}{2} \times \frac{213}{5}\end{aligned}$$

Simplify by cross-cancelling 25 and 5 (since  $25 \div 5 = 5$ ):

$$\begin{aligned}&= \frac{5 \times 213}{2} \\ &= \frac{1065}{2} \text{ km} \\ &= 532\frac{1}{2} \text{ km}\end{aligned}$$

**Answer:** The car can travel  **$532\frac{1}{2}$  km** (or **532.5 km**) in  $42\frac{3}{5}$  litres of petrol.

### Question 13

A graphic designer charges Rs.  $27\frac{3}{5}$  for each diagram. Find the amount he will charge if he designs 186 diagrams for a book.

**Solution:**

- Charge per diagram = Rs.  $27\frac{3}{5}$  = Rs.  $\frac{27 \times 5 + 3}{5}$  = Rs.  $\frac{138}{5}$ .
- Total number of diagrams = 186.

The total amount charged is the product of the rate and the number of diagrams:

$$\begin{aligned}\text{Total Charge} &= \text{Rate per diagram} \times \text{Number of diagrams} \\ &= \text{Rs. } \frac{138}{5} \times 186 \\ &= \text{Rs. } \frac{138 \times 186}{5} \\ &= \text{Rs. } \frac{25668}{5} \\ &= \text{Rs. } 5133\frac{3}{5}\end{aligned}$$



Note: Rs.  $5133\frac{3}{5}$  = Rs. 5133 and 60 paise (or Rs. 5133.60).

**Answer:** The graphic designer will charge **Rs. 5133.60**.

**Question 14**

If a cloth costs Rs.  $715\frac{1}{4}$  per metre, find the cost of  $3\frac{2}{5}$  metres of this cloth.

**Solution:**

- Cost of 1 metre of cloth = Rs.  $715\frac{1}{4}$  = Rs.  $\frac{715 \times 4 + 1}{4}$  = Rs.  $\frac{2861}{4}$ .
- Length of cloth purchased =  $3\frac{2}{5}$  metres =  $\frac{17}{5}$  metres.

The total cost of the cloth is the product of the rate and the length:

$$\begin{aligned}
 \text{Total Cost} &= \text{Rate per metre} \times \text{Total length} \\
 &= \text{Rs. } \frac{2861}{4} \times \frac{17}{5} \\
 &= \text{Rs. } \frac{2861 \times 17}{4 \times 5} \\
 &= \text{Rs. } \frac{48637}{20} \\
 &= \text{Rs. } 2431\frac{17}{20}
 \end{aligned}$$

Note: Rs.  $2431\frac{17}{20}$  = Rs. 2431.85.

**Answer:** The total cost of the cloth is **Rs. 2431.85**.

**Question 15**

Advertising in a magazine costs Rs.  $1472\frac{2}{5}$  per square inch. Find the cost of an advertisement of  $17\frac{1}{7}$  square inch.

**Solution:**

- Cost per square inch = Rs.  $1472\frac{2}{5}$  = Rs.  $\frac{1472 \times 5 + 2}{5}$  = Rs.  $\frac{7362}{5}$ .
- Area of the advertisement =  $17\frac{1}{7}$  sq. inches =  $\frac{17 \times 7 + 1}{7}$  =  $\frac{120}{7}$  sq. inches.

The total cost is the product of the rate and the area:

$$\begin{aligned}
 \text{Total Cost} &= \text{Rate per sq. inch} \times \text{Area} \\
 &= \text{Rs. } \frac{7362}{5} \times \frac{120}{7}
 \end{aligned}$$

Simplify by dividing 120 by 5 ( $120 \div 5 = 24$ ):

$$\begin{aligned}
 &= \text{Rs. } \frac{7362 \times 24}{7} \\
 &= \text{Rs. } \frac{176688}{7} \\
 &= \text{Rs. } 25241\frac{1}{7}
 \end{aligned}$$

Note: Rs.  $25241\frac{1}{7} \approx$  Rs. 25241.14.

**Answer:** The total cost of the advertisement is **Rs.  $25241\frac{1}{7}$**  (or approximately **Rs. 25241.14**).

**Question 16**

A car travelled from city A to city B with a uniform speed of  $52\frac{2}{7}$  km per hour. Find the distance between the two cities, if it took  $4\frac{3}{8}$  hours for the car to reach city B from city A.

**Solution:**

- Uniform speed of the car =  $52\frac{2}{7}$  km/h =  $\frac{52 \times 7 + 2}{7}$  km/h =  $\frac{366}{7}$  km/h.
- Time taken to travel =  $4\frac{3}{8}$  hours =  $\frac{4 \times 8 + 3}{8}$  hours =  $\frac{35}{8}$  hours.

The distance between the two cities is given by:

$$\begin{aligned}\text{Distance} &= \text{Speed} \times \text{Time} \\ &= \frac{366}{7} \times \frac{35}{8}\end{aligned}$$

Simplify by cross-cancelling 35 and 7 ( $35 \div 7 = 5$ ):

$$\begin{aligned}&= \frac{366 \times 5}{8} \\ &= \frac{1830}{8} \text{ km} \\ &= \frac{915}{4} \text{ km} \\ &= 228\frac{3}{4} \text{ km}\end{aligned}$$

Note:  $228\frac{3}{4}$  km = 228.75 km.

**Answer:** The distance between the two cities is  $228\frac{3}{4}$  km (or **228.75 km**).

**Question 17**

The length of a rectangular plot of land is  $29\frac{3}{7}$  m. If its breadth is  $12\frac{8}{11}$  m, find its area.

**Solution:**

- Length of the plot ( $l$ ) =  $29\frac{3}{7}$  m =  $\frac{29 \times 7 + 3}{7}$  m =  $\frac{206}{7}$  m.
- Breadth of the plot ( $b$ ) =  $12\frac{8}{11}$  m =  $\frac{12 \times 11 + 8}{11}$  m =  $\frac{140}{11}$  m.

The area of a rectangular plot is the product of its length and breadth:

$$\begin{aligned}\text{Area} &= \text{Length} \times \text{Breadth} \\ &= \frac{206}{7} \times \frac{140}{11}\end{aligned}$$

Simplify by cross-cancelling 140 and 7 ( $140 \div 7 = 20$ ):

$$\begin{aligned} &= \frac{206 \times 20}{11} \\ &= \frac{4120}{11} \text{ m}^2 \\ &= 374\frac{6}{11} \text{ m}^2 \end{aligned}$$

Note:  $374\frac{6}{11} \text{ m}^2 \approx 374.55 \text{ m}^2$ .

**Answer:** The area of the plot of land is  $374\frac{6}{11} \text{ m}^2$  (or approximately **374.55 m<sup>2</sup>**).